

**NATIONAL UNIVERSITY OF MODERN LANGUAGE
ISLAMABAD**

Lab task:-01,02,03



Submitted to:

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Submitted by:

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Subject: Operating System

Department: BSSE-4-Afternoon

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1: What is the file system used by the Linux?

The file system commonly used by Linux is called the Extended File System (ext). There have been several versions of the ext file system, with ext2, ext3, and ext4 being the most widely used. Each version has brought improvements in terms of performance, reliability, and features.

2: Name two types of boot loaders available.

Two types of boot loaders commonly used in Linux systems are:

GRUB (GRand Unified Bootloader): GRUB is one of the most popular boot loaders used in Linux distributions. It provides a flexible and feature-rich environment for booting multiple operating systems, configuring boot parameters, and troubleshooting boot-related issues. GRUB supports both BIOS-based (GRUB Legacy) and UEFI-based (GRUB 2) systems.

LILO (LIinux LOader): LILO was once a popular boot loader for Linux systems, but it has been largely replaced by GRUB. LILO is a simpler boot loader compared to GRUB and is primarily used in older systems or distributions that haven't adopted GRUB as the default boot loader. LILO has a configuration file named `lilo.conf` that defines the boot options and the location of the kernel image.

3: What are the names of partitions created for Linux?

Partitions created for Linux systems typically use a naming convention based on the device they are associated with. The most common naming scheme for partitions on Linux is based on the following pattern:

1. **Primary Partitions:** These are denoted by numbers (e.g., `/dev/sda1`, `/dev/sda2`, `/dev/sda3`, etc.). The first primary partition on the first disk is typically `/dev/sda1`, the second primary partition is `/dev/sda2`, and so on.
2. **Extended Partitions:** Historically, extended partitions were used to overcome the limit of four primary partitions per disk. Extended partitions do not directly contain file systems; instead, they serve as containers for logical partitions.
3. **Logical Partitions:** Logical partitions are partitions within extended partitions. They are also denoted by numbers but are typically numbered starting from 5 (e.g., `/dev/sda5`, `/dev/sda6`, `/dev/sda7`, etc.).

Lab No 02

task:- 01

```
zulqarnain@lenovo:~$ date
ت 09 مارچ 2024 و 16:30:41 PKT
zulqarnain@lenovo:~$ whoami
zulqarnain
zulqarnain@lenovo:~$ hostname
lenovo
zulqarnain@lenovo:~$ uname
Linux
zulqarnain@lenovo:~$ uptime
16:31:22 up 9 min, 1 user, load average: 0.43, 1.71, 1.26
zulqarnain@lenovo:~$
```

Task:- 02

```
zulqarnain@lenovo:~$ ls
Desktop Documents Downloads Music Pictures Public snap Templates Videos
zulqarnain@lenovo:~$ cat> hello.txt
Hello world!
this is a text file.
zulqarnain@lenovo:~$ ls
Desktop Documents Downloads hello.txt Music Pictures Public snap Templates Videos
zulqarnain@lenovo:~$ cat hello.txt
Hello world!
this is a text file.
zulqarnain@lenovo:~$ cat hello.txt
Hello world!
this is a text file.
```

Task:- 03

```
zulqarnain@lenovo:~$ cat> secret-of-the-univers
hi
I
am
a
BSse
student

zulqarnain@lenovo:~$ l
Desktop/   Downloads/ Music/   Public/   snap/   Videos/
Documents/ hello.txt Pictures/ secret-of-the-univers Templates/
zulqarnain@lenovo:~$ ls
Desktop Documents Downloads hello.txt Music Pictures Public secret-of-the-univers snap Templates Videos
zulqarnain@lenovo:~$ cat secret-of-the-univers
hi
I
am
a
BSse
student
```

Task:- 04

```
zulqarnain@lenovo:~$ cp secret-of-the-univers answer.txt
zulqarnain@lenovo:~$ cat answer.txt
hi
I
am
a
BSse
student

zulqarnain@lenovo:~$ cp hello.txt answer.txt
zulqarnain@lenovo:~$ cat answer.txt
Hello world
this is a text file.
zulqarnain@lenovo:~$ rm hello.txt
zulqarnain@lenovo:~$ ls
answer.txt Desktop Documents Downloads Music Pictures Public secret-of-the-univers snap Templates Videos
zulqarnain@lenovo:~$ mv answer.txt message
zulqarnain@lenovo:~$ ls
Desktop Documents Downloads message Music Pictures Public secret-of-the-univers snap Templates Videos
zulqarnain@lenovo:~$ rm missing
rm: cannot remove 'missing': No such file or directory
zulqarnain@lenovo:~$ cp secret-of-the-univers
cp: missing destination file operand after 'secret-of-the-univers'
Try 'cp --help' for more information.
zulqarnain@lenovo:~$
```

Lab No 03

task:- 01

```
zulqarnain@lenovo:/$ pwd
/
zulqarnain@lenovo:/$ cd /home/zulqarnain
zulqarnain@lenovo:~$ pwd
/home/zulqarnain
zulqarnain@lenovo:~$ cd ..
zulqarnain@lenovo:/home$ cd ..
zulqarnain@lenovo:/$ pwd
/
zulqarnain@lenovo:/$ ls
bin  boot  cdrom  dev  etc  home  lib  lib32  lib64  libx32  lost+found  media  mnt  opt  proc  root  run  sbin  snap  srv  swapfile  sys  tmp  usr  var
zulqarnain@lenovo:/$ cd home
zulqarnain@lenovo:/home$ ls
zulqarnain
zulqarnain@lenovo:/home$
```

task:- 02

```
zulqarnain@lenovo:~$ cat> shakespeare.txt
Shal i compare thee to a summer's day?
thou art more lovely and more temperate
zulqarnain@lenovo:~$ mv shakespeare.txt sonnet-18.txt
zulqarnain@lenovo:~$ mkdir poetry
zulqarnain@lenovo:~$ mv poem poetry
mv: cannot stat 'poem': No such file or directory
```

task:-3

```
zulqarnain@lenovo:/$ ls
bin boot cdrom dev etc home lib lib32 lib64 libx32 lost+found media mnt opt proc root run sbin snap srv swapfile sys tmp usr var
zulqarnain@lenovo:/$ cd usr/share
zulqarnain@lenovo:/usr/share$ ls
accountsservice dns gnome-remote-desktop keyrings open-vm-tools source-highlight
aclocal dnsmasq-base gnome-session language-selector openvpn speech-dispatcher
acpi-support doc gnome-settings-daemon language-support orca ssl-cert
adduser doc-base gnome-shell language-tools org.gnome.Characters system-config-printer
aisleriot dpkg gnome-system-monitor libc-bin os-prober systemd
alsa drirc.d gnome-todo libdebuginfod-common p11-kit tabset
alsa-base emacs gnome-video-effects libdrm PackageKit tcltk
alsa-card-profile emacsen-common gnupg libbexttextcat perl terminfo
appdata enchant-2 goa-1.0 libgnomekbd pam themes
applications eog grilo-plugins libgweather pam-configs thumbnails
apport evince grub libinput perl totem
apturl evolution-data-server grub2 liblangtag perl5 tracker3
aspell file grub-gfxpayload-lists liblouis pipewire transmission
avahi file-roller gst-plugins-base liblouisutdm1 pixmaps ubuntu
backgrounds fish gstreamer-1.0 libmysofa pkgconf ubuntu-drivers-common
base-files fontconfig gtk-4.0 libreoffice plymouth ubuntu-release-upgrader
base-passwd fonts gtk-doc librevenge pnm2ppa ubuntu-wayland
bash-completion fonts-droid-fallback glib-2.0 libwacom pockeetsphinx ucf
binfmts fonts-sil-abyssinica glib-2.0 libwacom pockeetsphinx ucf
branding fonts-sil-padauk glib-2.0 libwacom pockeetsphinx ucf
brltty foo2qpd1 glib-2.0 libwacom pockeetsphinx ucf
bug foo2zjs glib-2.0 libwacom pockeetsphinx ucf
build-essential fwupd gamemode gcc locales python3 python-apt vala
ca-certificates gconf GConf help help-langpack man man-db readline vin
cnae gconv GConf help help-langpack man man-db readline vin
color gdb gdm hunspell hyphen i18n ibus icons mfx mime seahorse xsessions
colord gdm hunspell hyphen i18n ibus icons mfx mime seahorse xsessions
common-licenses gedit gettext i18n ibus icons mfx mime seahorse xsessions
consolefonts ghostscript i18n ibus icons mfx mime seahorse xsessions
console-setup glib-1.0 glade ibus-table in-config indicators info initramfs-tools ModemManager session-migration yelp-xsl
consoletrans glib-2.0 glade ibus-table in-config indicators info initramfs-tools ModemManager session-migration yelp-xsl
cups glib-2.0 glade ibus-table in-config indicators info initramfs-tools ModemManager session-migration yelp-xsl
dbus-1 glib-2.0 glade ibus-table in-config indicators info initramfs-tools ModemManager session-migration yelp-xsl
dconf glib-2.0 glade ibus-table in-config indicators info initramfs-tools ModemManager session-migration yelp-xsl
debconf glib-2.0 glade ibus-table in-config indicators info initramfs-tools ModemManager session-migration yelp-xsl
debhelper glib-2.0 glade ibus-table in-config indicators info initramfs-tools ModemManager session-migration yelp-xsl
debiutils glib-2.0 glade ibus-table in-config indicators info initramfs-tools ModemManager session-migration yelp-xsl
defaults glib-2.0 glade ibus-table in-config indicators info initramfs-tools ModemManager session-migration yelp-xsl
```

```
zulqarnain@lenovo:/usr/share$ cd ..
zulqarnain@lenovo:/usr$ ls
bin games include lib lib32 lib64 libexec libx32 local sbin share src
zulqarnain@lenovo:/usr$ cd ..
zulqarnain@lenovo:/$ ls
bin boot cdrom dev etc home lib lib32 lib64 libx32 lost+found media mnt opt proc root run sbin snap srv swapfile sys tmp usr var
zulqarnain@lenovo:/$ echo ~
/home/zulqarnain
zulqarnain@lenovo:/$ cat ~/message /
Hello world
this is a text file.
cat: /: Is a directory
zulqarnain@lenovo:/$
```

task:- 04

```
zulqarnain@lenovo:/$ hostname
lenovo
zulqarnain@lenovo:/$ locate hostname
/etc/hostname
/snap/core22/1122/etc/hostname
/snap/core22/1122/etc/writable/hostname
/snap/core22/1122/usr/bin/hostname
/snap/core22/1122/usr/bin/hostnamectl
/snap/core22/1122/usr/lib/python3/dist-packages/cloudinit/config/cc_set_hostname.py
/snap/core22/1122/usr/lib/python3/dist-packages/cloudinit/config/cc_update_hostname.py
/snap/core22/1122/usr/lib/python3/dist-packages/cloudinit/config/__pycache__/cc_set_hostname.cpython-310.pyc
/snap/core22/1122/usr/lib/python3/dist-packages/cloudinit/config/__pycache__/cc_update_hostname.cpython-310.pyc
/snap/core22/1122/usr/lib/python3/dist-packages/cloudinit/distros/parsers/hostname.py
/snap/core22/1122/usr/lib/python3/dist-packages/cloudinit/distros/parsers/__pycache__/hostname.cpython-310.pyc
/snap/core22/1122/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname
/snap/core22/1122/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/__init__.py
/snap/core22/1122/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/__pycache__
/snap/core22/1122/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/_implementation.py
/snap/core22/1122/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/__pycache__/__init__.cpython-310.pyc
/snap/core22/1122/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/__pycache__/implementation.cpython-310.pyc
/snap/core22/1122/usr/lib/systemd/systemd-hostnamed
/snap/core22/1122/usr/lib/systemd/system/dbus-org.freedesktop.hostname1.service
/snap/core22/1122/usr/lib/systemd/system/systemd-hostnamed.service
/snap/core22/1122/usr/share/bash-completion/completions/hostname
/snap/core22/1122/usr/share/bash-completion/completions/hostnamectl
/snap/core22/1122/usr/share/dbus-1/system-services/org.freedesktop.hostname1.service
/snap/core22/1122/usr/share/dbus-1/system.d/org.freedesktop.hostname1.conf
/snap/core22/1122/usr/share/doc/hostname
/snap/core22/1122/usr/share/doc/hostname/changelog.gz
/snap/core22/1122/usr/share/doc/hostname/copyright
/snap/core22/1122/usr/share/polkit-1/actions/org.freedesktop.hostname1.policy
/snap/core22/1122/usr/share/zsh/vendor-completions/_hostnamectl
/snap/gnome-42-2204/141/usr/lib/python3/dist-packages/pip/_vendor/urllib3/util/ssl_match_hostname.py
/snap/gnome-42-2204/141/usr/lib/x86_64-linux-gnu/gettext/hostname
/usr/bin/hostname
/usr/bin/hostnamectl
/usr/bin/nmtui-hostname
/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname
/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/__init__.py
/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/__pycache__
/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/_implementation.py
/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/__pycache__/__init__.cpython-310.pyc
/usr/lib/python3/dist-packages/urllib3/packages/ssl_match_hostname/__pycache__/implementation.cpython-310.pyc
/usr/lib/systemd/systemd-hostnamed
/usr/lib/systemd/system/dbus-org.freedesktop.hostname1.service
/usr/lib/systemd/system/systemd-hostnamed.service
/usr/share/bash-completion/completions/hostname
/usr/share/bash-completion/completions/hostnamectl
/usr/share/dbus-1/system-services/org.freedesktop.hostname1.service
/usr/share/dbus-1/system.d/org.freedesktop.hostname1.conf
/usr/share/doc/hostname
/usr/share/doc/hostname/changelog.gz
/usr/share/doc/hostname/copyright
/usr/share/man/man1/hostname.1.gz
/usr/share/man/man1/hostnamectl.1.gz
```

```
zulqarnain@lenovo:/$ cd user/bin/hostname
bash: cd: user/bin/hostname: No such file or directory
zulqarnain@lenovo:/$ cd /user/bin/hostname
bash: cd: /user/bin/hostname: No such file or directory
zulqarnain@lenovo:/$ cd /usr/bin/hostname
bash: cd: /usr/bin/hostname: Not a directory
zulqarnain@lenovo:/$ cd usr/bin/hostname
bash: cd: usr/bin/hostname: Not a directory
zulqarnain@lenovo:/$
```

task:- 05

```
zulqarnain@lenovo:~$ echo *
Desktop Documents Downloads message Music Pictures poetry Public secret-of-the-univers snap sonnet-18.txt Templates Videos
zulqarnain@lenovo:~$ echo '*'
*
zulqarnain@lenovo:~$ ls
Desktop Documents Downloads message Music Pictures poetry Public secret-of-the-univers snap sonnet-18.txt Templates Videos
zulqarnain@lenovo:~$ cp poeam.txt sonnet-18.txt
cp: cannot stat 'poeam.txt': No such file or directory
zulqarnain@lenovo:~$ cp sonnet-18.txt poeam.txt
zulqarnain@lenovo:~$ ls
Desktop Documents Downloads message Music Pictures poeam.txt poetry Public secret-of-the-univers snap sonnet-18.txt Templates Videos
zulqarnain@lenovo:~$ echo *
Desktop Documents Downloads message Music Pictures poeam.txt poetry Public secret-of-the-univers snap sonnet-18.txt Templates Videos
zulqarnain@lenovo:~$ echo '*'
*
zulqarnain@lenovo:~$ cat *.txt
Shal i compare thee to a summer's day?
thou art more lovely and more temperate
Shal i compare thee to a summer's day?
thou art more lovely and more temperate
zulqarnain@lenovo:~$ rmdir poetry
zulqarnain@lenovo:~$ ls
Desktop Documents Downloads message Music Pictures poeam.txt Public secret-of-the-univers snap sonnet-18.txt Templates Videos
zulqarnain@lenovo:~$
```